What drives WTI Oil Prices?

Evidence from CFTC (Commodity Futures Trading Commission) Disaggregated Data and OLS Regression Analysis.

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The "Financialised" Oil Market

- **Market Structure:** Not just physical supply/demand. The market is split into Spot (Physical) and Derivatives (Futures/Options).
- **The "Financialization" Thesis:** Since the mid-2000s, oil has become an asset class.
- **New Players:** Hedge Funds, CTAs, and Swap Dealers trade oil not to use it, but for portfolio exposure.
- **The Hypothesis:** Their financial flows (buying/selling futures) can decouple price from physical fundamentals.

100M
PHYSICAL BARRELS / DAY
CONSUMPTION

vs

6B
FINANCIAL BARRELS / DAY
TRADING VOLUME

Physical vs. Financial

When "paper barrels" outnumber "wet barrels" 60:1, who sets the price?

Market Structure: The Players

WTI is a producer-dominated market. Price formation is a conflict between three distinct groups:

Managed Money ("Fast Money")
Hedge funds & CTAs. The only group consistently generating Alpha. They set the price on the margin.

Producers ("The Shorts")
Sell futures to hedge extraction. Their selling pressure creates the risk premium.

Swap Dealers
Intermediaries facilitating trade, typically taking the other side of producers.

Question

Given the 60x leverage of financial vs. physical markets

What truly drives the price of oil today?

Macro Supply & Demand?

OR

Trader Positioning?

Why OLS Regression?

I' used a multi variate OLS regression to see the relationships between the dependent variable WTI prices and 4 independent variables (HF, SD…)

Objective

Quantify the relationship between weekly price changes and changes in trader positions.

Transparency

OLS is standard, interpretable, and provides direct coefficients, t-stats, and confidence intervals.

Multi-Factor

We control for 4 distinct **CFTC*** categories simultaneously to isolate which group drives the market.

$$\Delta Oil_t = \alpha + \beta_{HF}\Delta HF_t + \beta_{Swap}\,\Delta Swap_t + \beta_{Prod}\Delta Prod_t + \beta_{Other}\Delta Other_t + \varepsilon_t$$

*Commodity Futures Trading Commission

Data Sources

1. CFTC Disaggregated Report

`File: Disaggregated_Combined.xlsx`

- **Frequency:** Weekly (Tuesday)
- **Target:** WTI-PHYSICAL (NYMEX)
- **Variables:** Net Position (Long - Short) for:
  - Managed Money (Hedge Funds)
  - Producer/Merchant
  - Swap Dealers
  - Other Reportables

2. Market Price Data

`Source: Yahoo Finance (yfinance)`

- **Ticker:** CL=F (Front-month Futures)
- **Method:** Adjusted Close Prices
- **Alignment:** Merged to CFTC report dates (Tuesdays) using backward fill.

Step 1: Load & Filter CFTC Data

```
import pandas as pd # Load Excel Sheet df =
pd.read_excel(FILE_PATH, sheet_name="Data")
df["date"] =
pd.to_datetime(df["Report_Date_as_YYYY_MM_DD"]) #
Filter for WTI only wti =
df[df["Market_and_Exchange_Names"] ==
MARKET_NAME].copy() wti = wti[wti["date"] ≥
START_DATE].sort_values("date")
```

Logic:

We read the raw CFTC file and isolate the specific market contract. We also standardize the date format to allow for merging later.

Step 2: Build Net Positions

The Formula:

```
Net = Long - Short
```

We care about the **Net Economic Exposure**. Are they bullish (Net > 0) or bearish (Net < 0)?

```
# Calculate Net Positions wti["hf_net"] =
wti["M_Money_Positions_Long_All"] - \
wti["M_Money_Positions_Short_All"] wti["prod_net"]
= wti["Prod_Merc_Positions_Long_All"] - \
wti["Prod_Merc_Positions_Short_All"] # Repeat for
Swap & Other ...
```

Step 3: Merge Prices

```
# Download Price Data oil_raw = yf.download("CL=F",
start=START_DATE) # Merge 'As Of' (Backward
direction) merged = pd.merge_asof(
wti.sort_values("date"), oil.sort_values("date"),
on="date", direction="backward" ).dropna()
```

merge_asof

This is crucial. CFTC reports are weekly (Tuesday). Prices are daily.

merge_asof finds the nearest price **on or before** the report date to ensure causality (we don't use future prices).

Step 4: Stationarity & OLS

```
# 1. First Differences (Flows) merged["d_oil"] = merged["oil_price"].diff() merged["d_hf"] =
merged["hf_net"].diff() # 2. Define Variables Y = reg_data["d_oil"] X = sm.add_constant(reg_data[["d_hf",
"d_swap", "d_prod"]]) # 3. Fit with Robust Errors model = sm.OLS(Y, X).fit(cov_type="HC1")
```

We regress **Price Changes** on **Position Changes (Flows)**.

Scope: Why Only WTI-NYMEX?

The CFTC dataset contains dozens of energy contracts (Brent, WCS, RBOB, Heating Oil). We filter strictly for:

```
"WTI-PHYSICAL - NEW YORK MERCANTILE EXCHANGE"
```

- **Benchmark Status:** The global reference for light sweet crude.
- **Liquidity:** The deepest pool of speculative money.
- **Clarity:** Mixing contracts (e.g., Brent vs WTI, futures and options) introduces regional spread noise and multicollinearity.

Regression Results

MODEL SPECIFICATION

$$\Delta \mathrm{Oil}t = \alpha + \beta 1 \Delta \mathrm{MM}t + \beta 2 \Delta \mathrm{Prod}t + \beta 3 \Delta \mathrm{Swap}t + \beta 4 \Delta \mathrm{Other}t + \varepsilon t$$

Variable	Coef	Std err	t-stat	p-value
const	-0.0668	0.1668	-0.40	0.689
d-hf	0.000209	0.000031	6.77	1.3e-11
d_swap	0.000104	0.000050	2.06	0.039
d_prod	0.000136	0.000030	4.49	7.2e-06
d_other	0.000187	0.000031	5.96	2.6e-09

Intercept (const): Approximately 0. When position changes are zero, the expected weekly oil price change is basically zero

All four flow variables are positive and strongly significant at 5% with d_swap being the weakest (t ≈ 2.06, just above the 1.977 critical value)

Interpretation of magnitudes:

d_hf = 0.000209: if hedge-fund net positions increase by 10,000 contracts in a week, expected **ΔWTI ≈ 0.000209×10,000 ≈ 2.09 USD/bbl**

R^2, F-test and Overall Fit

Model Performance & Residual Diagnostics

EXPLANATORY POWER

$R^2 = 0.539$

About **54%** of the variance in weekly oil price changes is explained by futures position flows.

Adjusted R^2 = 0.525

JOINT SIGNIFICANCE

F-static = 63.57

p ≈ 1.9e-30

Strongly reject the null hypothesis. Together, these flows clearly matter.

BENCHMARK COMPARISON

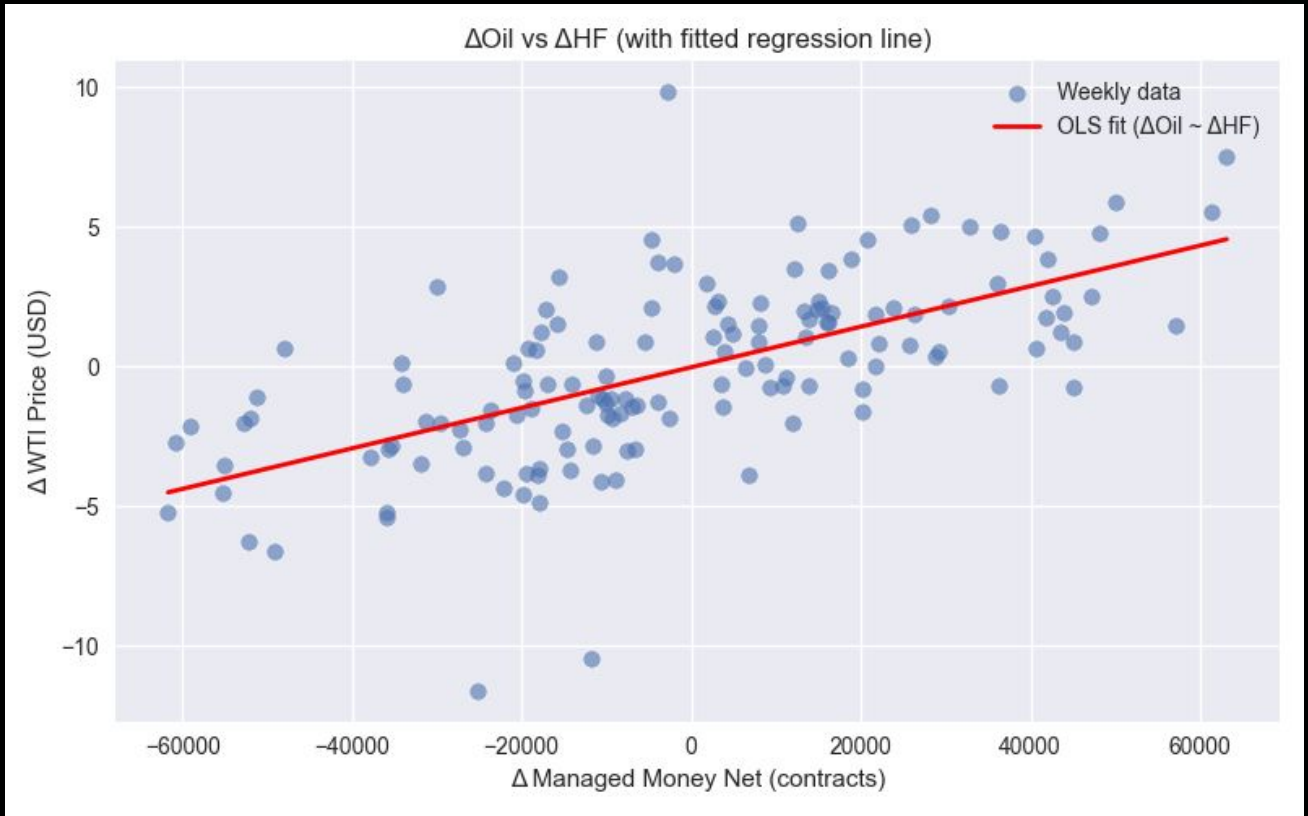
vs 0.14

This CFTC model explains much more variance than the Macro Regression model ($R^2 \approx 0.14$).

>_ Residual Diagnostics Output

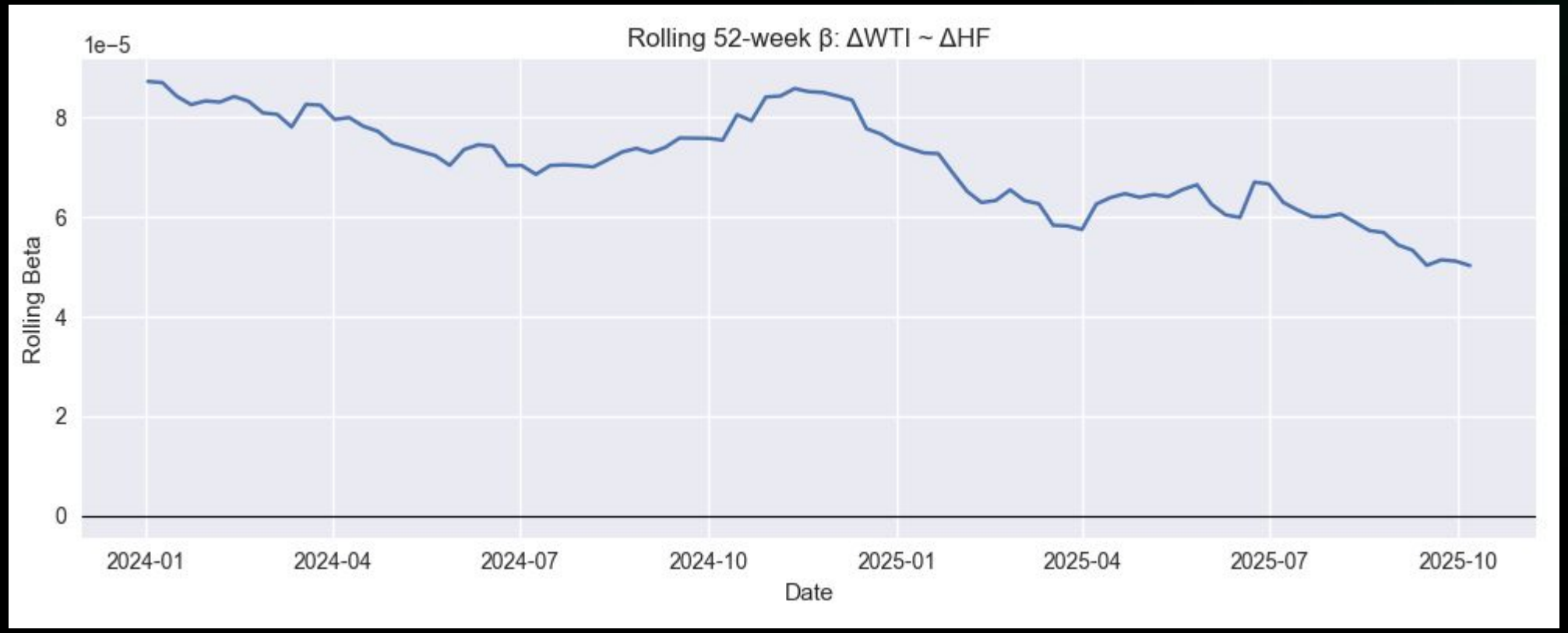
Omnibus: 39.200	Durbin-Watson: 2.431	Prob(Omnibus): 0.000	Jarque-Bera (JB): 247.975
Skew: -0.717	Prob(JB): 1.42e-54	Kurtosis: 9.267	Cond. No. 3.22e+04

Diagnostic 1: Scatter Plot



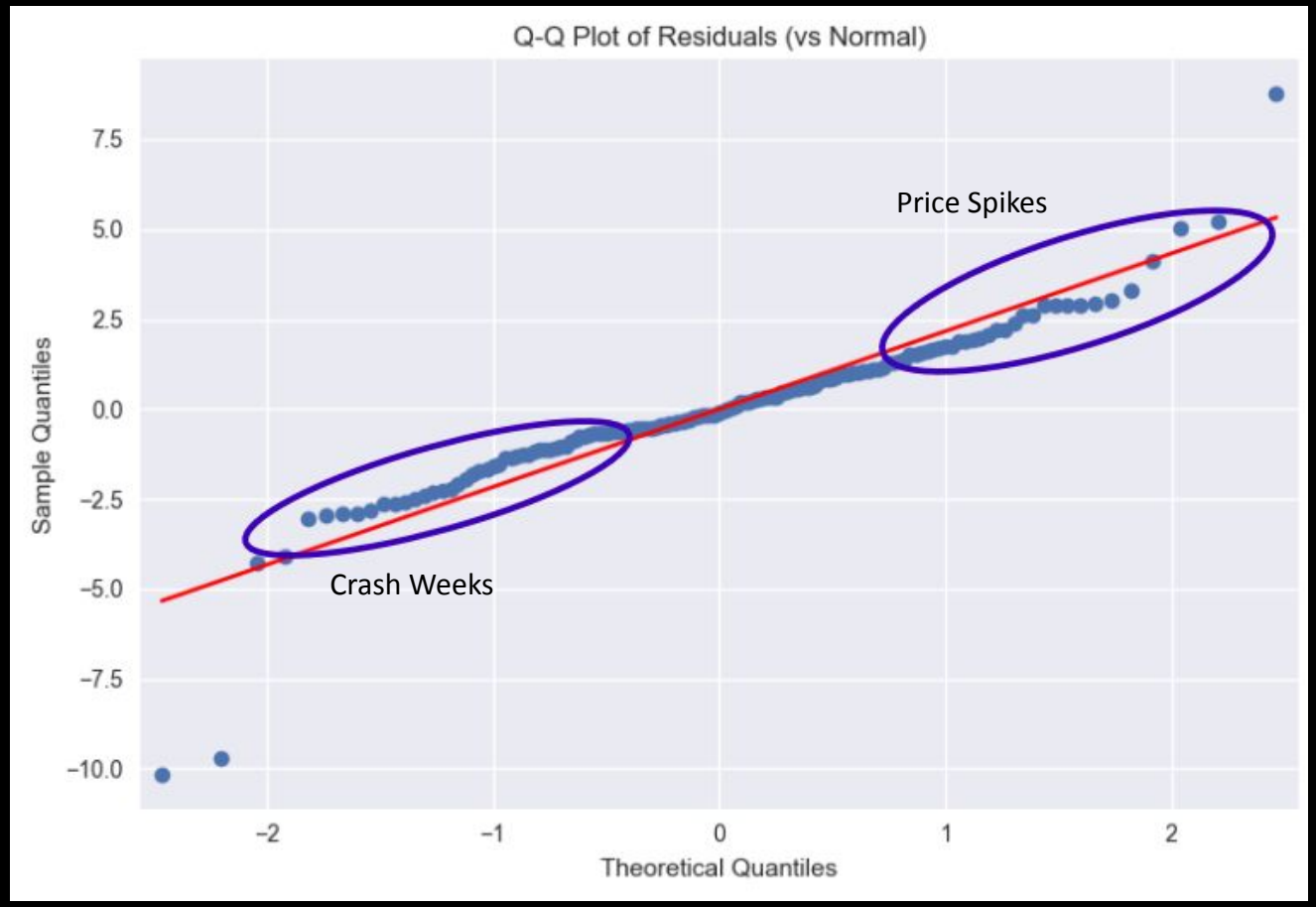
ΔOil vs ΔHF: Strong positive correlation.

Diagnostic 2: Rolling 52-week Beta



Rolling beta remains positive but gradually declines, indicating changing influence of hedge-fund flows.

Diagnostic 4: Q-Q Plot

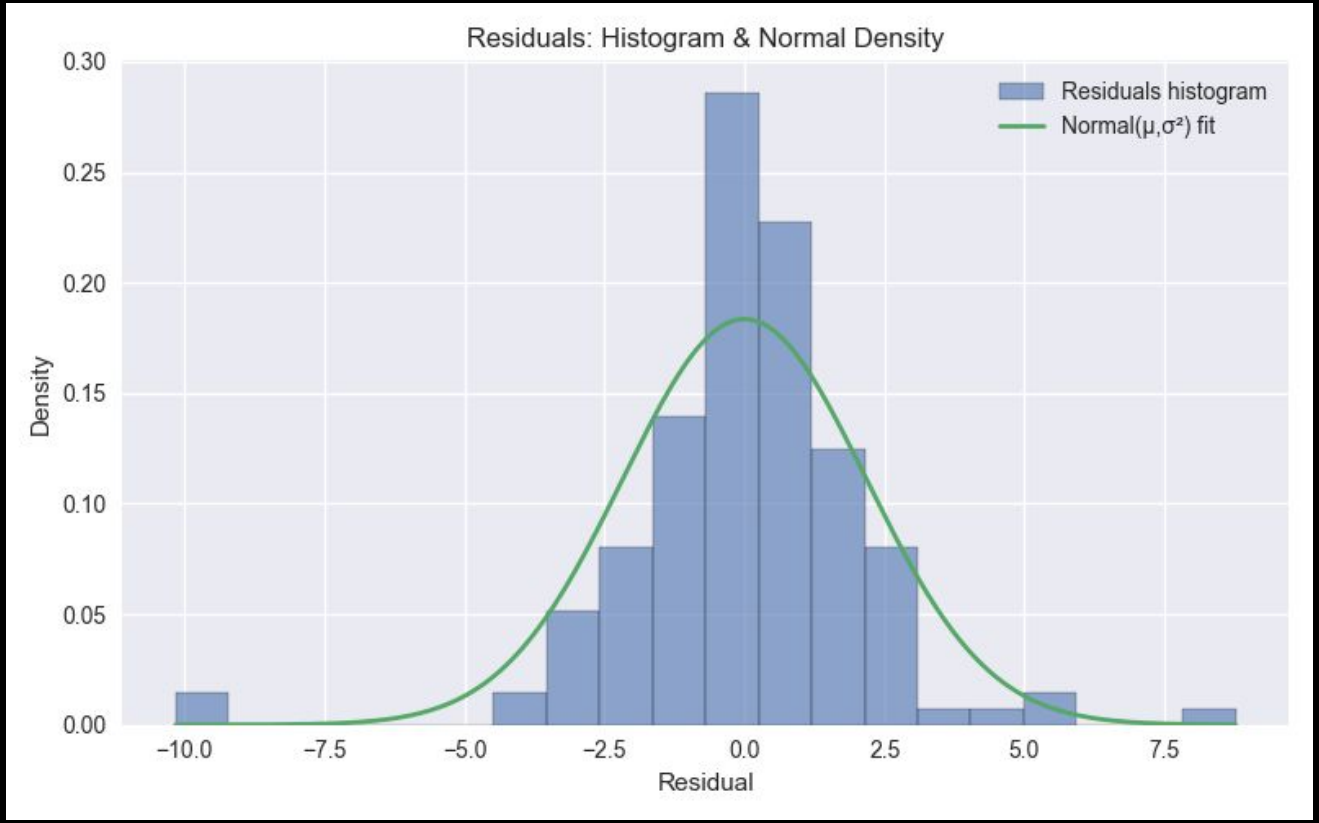


- Middle: close to normal
- Tails: strong deviations
- Evidence of fat tails / non-normality

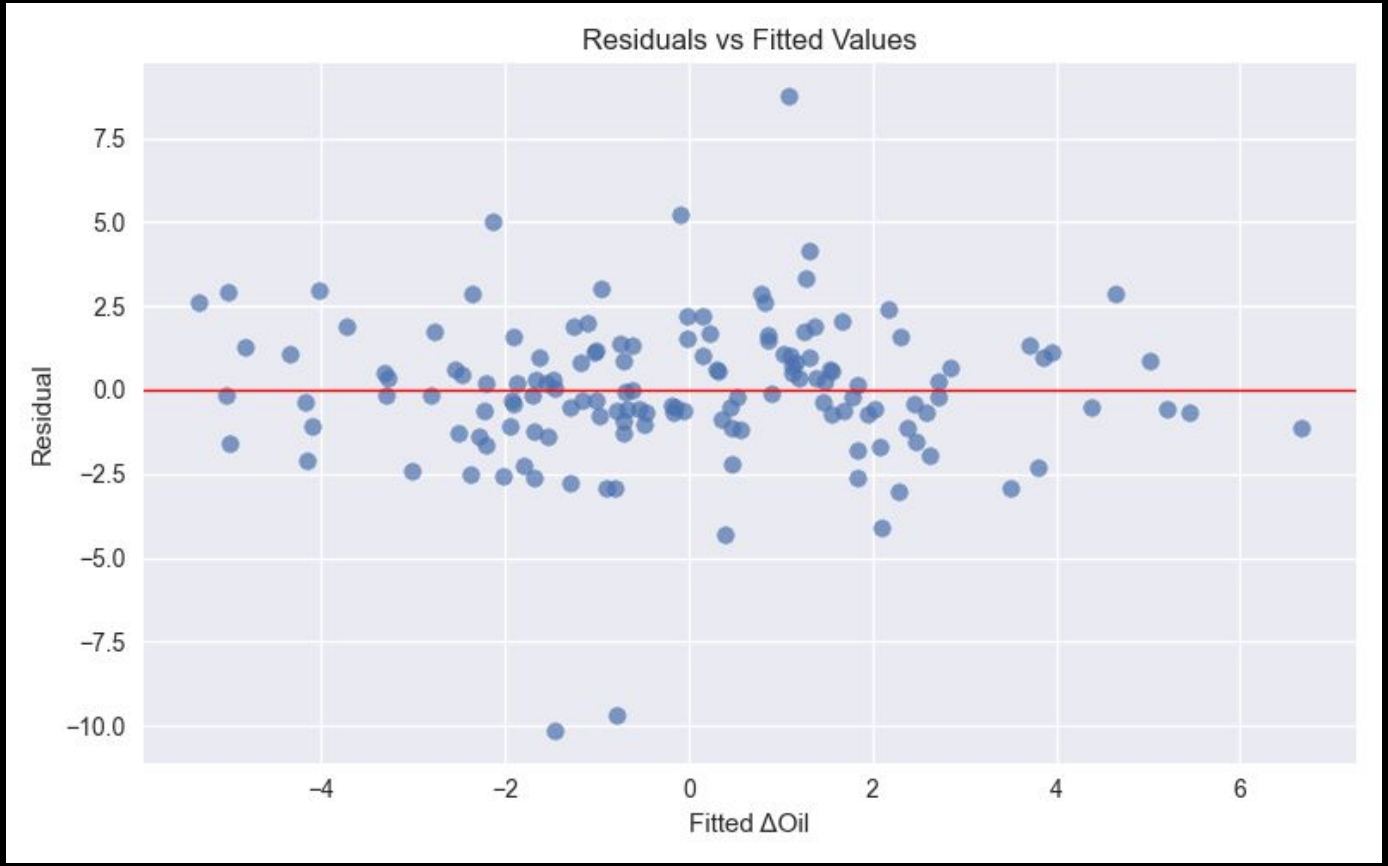
Diagnostic 3: Residuals Distribution vs Normal Fit

Non-Normality

Residuals show fat tails: extreme oil-price movements are more frequent than OLS assumes.

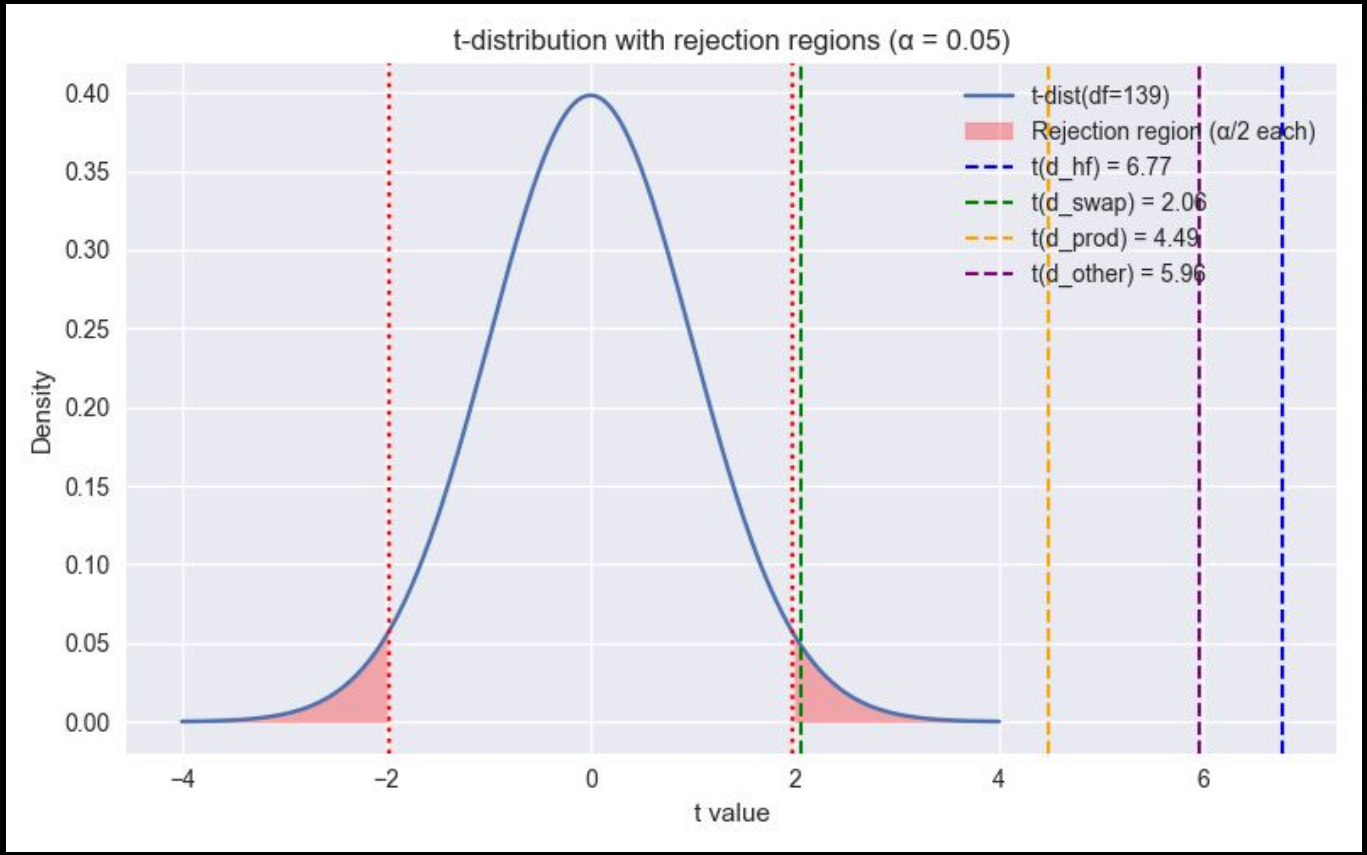


Diagnostic 5: Residual vs Fitted



No systematic curve observed. Linear specification is adequate.

Hypothesis Testing: T-Stats



The calculated t-statistics fall far into the rejection region (> 1.96).

We reject the Null Hypothesis of "No Effect".

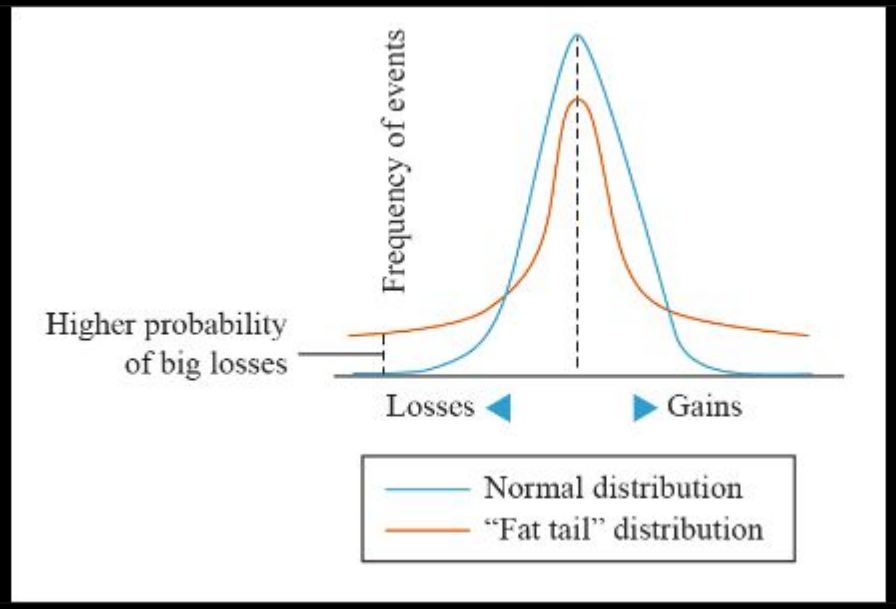
Why HC1 Robust Errors?

The Problem: Heteroscedasticity

- Assumption Violation: Standard OLS assumes **homoscedasticity**, meaning the variance of error terms remains constant.
- Time Series Reality: Financial data (like WTI weekly price changes) often exhibits heteroscedasticity, where **volatility changes over time** (volatility clustering).
- Consequence: Because the assumption is violated, the calculated standard errors are biased. This renders the resulting **t-statistics and p-values unreliable for hypothesis testing.**

The Solution: HC1

- Correction Mechanism: HC1 (**Heteroscedasticity Consistent**) estimators correct this bias without changing the coefficient estimates.
- Individual Variance: Instead of assuming a constant global variance, HC1 utilizes the **variance of each specific error term** (squared residuals).
- Result: By weighing errors based on their individual variance, **HC1 provides valid standard errors**, restoring the accuracy of t-tests and p-values.



Visualizing Fat Tails (Kurtosis)

Flows vs. Macro Model

We ran a second regression using standard macro factors:

- S&P 500
- US Dollar Index (DXY)
- VIX
- US 10Y Yield

Explanatory Power (R^2)

CFTC Flows Model

Macro Factor Model

"How this regression is useful in real business contexts ?"

Recognize when price moves are driven by speculators, not supply-demand.
Hedge inflation-themed portfolios more effectively using flow indicators.
Detect flow-driven mispricing to exploit short-term arbitrage opportunities.
Build macro-commodity models linking financial flows to oil asset returns.
Improve budgeting accuracy by adding a "financial flow risk factor" on top of supply-demand models.
Forecast short-term fuel cost movements to lock in procurement rates.
Use ΔHF / ΔSD coefficients to forecast optimal hedge execution windows.
Monitor aggressive selling by commercial producers → signals risk of price declines.
Identify when hedge funds are pushing futures higher → signal to hedge production earlier.
Improve hedging timing by reacting swiftly to extreme Hedge Fund/Swap Dealer position changes.
Anticipate volatility spikes driven by position imbalances rather than fundamentals.
Detect whether speculative flows (hedge funds) are creating artificial short-term price pressure.

STRATEGIC IMPLICATIONS

Applying Financial Flow Analysis to Industry Operations

MARKET RISK MANAGEMENT

ENERGY & TRADING FIRMS

- Detect whether speculative flows (hedge funds) are creating artificial short-term price pressure.
- Anticipate volatility spikes driven by position imbalances rather than fundamentals.
- Improve hedging timing by reacting swiftly to extreme Hedge Fund/Swap Dealer position changes.

HEDGING OPTIMIZATION

OIL PRODUCERS & REFINERS

- Identify when hedge funds are pushing futures higher → signal to hedge production earlier.
- Monitor aggressive selling by commercial producers → signals risk of price declines.
- Use ΔHF / ΔSD coefficients to forecast optimal hedge execution windows.

SUPPLY CHAIN & PROCUREMENT

AIRLINES, SHIPPING, CHEMICALS

- Forecast short-term fuel cost movements to lock in procurement rates.
- Improve budgeting accuracy by adding a "financial flow risk factor" on top of supply-demand models.
- Recognize when price moves are driven by speculators, not supply-demand.

INVESTOR & PORTFOLIO MGMT

ASSET MANAGERS & QUANTS

- Build macro-commodity models linking financial flows to oil asset returns.
- Detect flow-driven mispricing to exploit short-term arbitrage opportunities.
- Hedge inflation-themed portfolios more effectively.

🏁 Conclusion

- **Financialization Confirmed:** Futures flows explain ~**54%** of weekly price changes, far more than macro factors (**14%**).
- **Key Drivers:** All 4 CFTC categories are statistically significant. Hedge Funds (Managed Money) are the momentum engine.
- **Methodological Rigor:** Financial data requires Robust Standard Errors (HC1) to handle fat tails.
- **Verdict:** Who is buying/selling "paper barrels" matters more than general macro sentiment for short-term WTI pricing.